Predictive Modeling of Smoking Status Using Bio-Signals and Machine Learning Approaches

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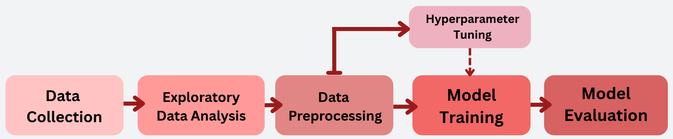
Abstract

Smoking is a leading cause of death and disability worldwide. This study utilized medical and lifestyle data to develop a smoker status prediction model using machine learning techniques. Logistic Regression, Random Forest, XGBoost, K-Nearest Neighbors (KNN), and LightGBM were applied and compared to evaluate their performance.

Introduction

Smoking is one of the main causes of serious diseases and early death around the world. A person's smoking status is very important for preventive healthcare because smoking can cause many health problems such as heart disease, cancer, and lung issues. **The World Health Organization (WHO) estimates that by 2030, about 10 million people will die every year because of smoking**, which is around 9% of the world's population. Even though the dangers of smoking are well-known, tobacco companies still spend more than $4 billion each year on advertisements. Because of this, it is important to find ways to identify smokers early. This study builds a machine learning model to predict smoking status based on various health indicators.

Workflow



This workflow demonstrates a machine learning pipeline that processes physiological bio-signals for accurate smoking status classification.

Methods

This study builds a prediction model to classify smoking status using bio-signal data and machine learning methods.

Data Collection

We used data from Kaggle. The dataset contains

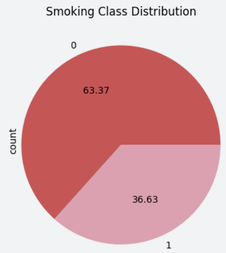
23 Features

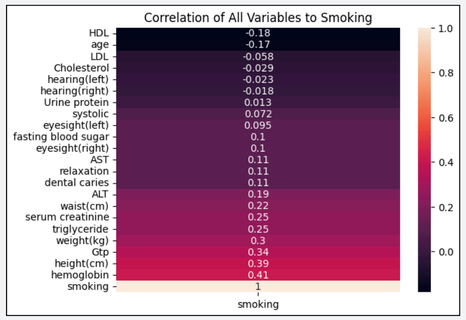
38,984 Rows

It includes different body measurements such as blood pressure, blood sugar, cholesterol, eyesight, hearing, waist size, hemogoblin, and other health signals.

Exploratory Data Analysis

Data distributions were explored to check for missing values, imbalances, and correlations between features. This step helps to better understand the data before model building.





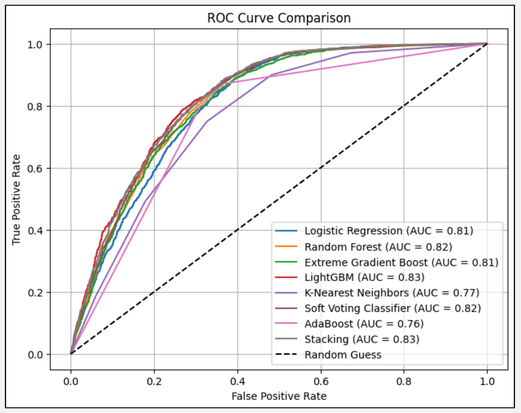
Data Preprocessing

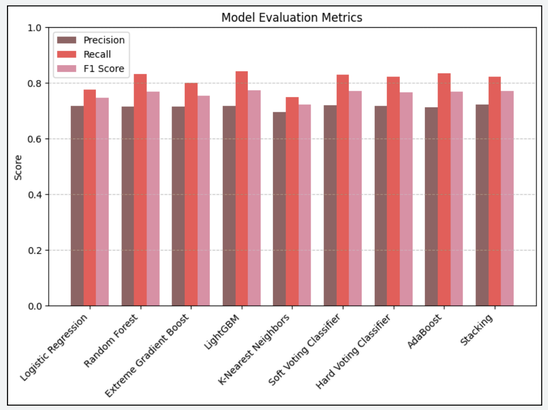
We handled outliers and duplicate rows, encoded categorical variables, and scaling data. To address data imbalance, we applied downsampling. The dataset was then split into 80% for training and 20% for testing.

Model Training

Several machine learning models: Logistic Regression, Random Forest, XGBoost, LightGBM, K-Nearest Neighbors (KNN) were used. These models were selected because they offer different strengths for classification tasks.

Model Evaluation





Analysis

LightGBM achieved the highest performance with an ROC-AUC of **0.83** and a recall of **84% for smokers**, indicating its strong ability to correctly identify most smokers. The **F1-score of 0.78 in Smoking Class** reflects a balanced trade-off between precision and recall, making it a reliable choice in scenarios where both false positives and false negatives matter. In early health screening, high recall is essential to minimize missed cases, though improving precision would help reduce false alarms. These results show the model's potential in supporting preventive healthcare through simple health indicators.

Conclusions

The models successfully classified smoking status based on bio-signal data. Ensemble models like LightGBM and Stacking Classifier showed higher performance. This method may help support healthcare providers in identifying smoking behavior using simple health data.

Future Work

Future studies can use larger and more diverse datasets, include additional features such as genetic or behavioral data, and explore more deep learning models for better prediction accuracy.

References

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